Roll No. | | | | | | | |

Paper Code: OCE-351

M.Tech. (Geo-Informatics)

**Mid Semester Examination 2022**

*III Semester*

**Cost Management of Engineering Projects**

Time: 1:30 Hours

MM: 50

**Note:**

(i) All questions are compulsory.

(ii) Answer any one sub questions among a & b in each main question.

(iii) Total marks for each main question are ten.

Q1

(1x10 = 10 Marks)

a Describe Opportunity cost, Incremental cost, Differential cost and Relevant cost.

b Explain in detail Cost concepts in decision-making. CO1

Q2

(1x10 = 10 Marks)

a Explain Creation of a Database for operational control?

b Describe the objectives of a costing system? CO1

Q3

(1x10 = 10 Marks)

a Describe project execution as conglomeration of technical and nontechnical activities.

b Describe briefly pre-project execution and its importance in project work. CO2

Q4

(1x10 = 10 Marks)

a Describe briefly pre project execution main clearances and its impact on project.

b Explain in detail the importance of project site and project documentation phases. CO2

Q5

(1x10 = 10 Marks)

a Explain in detail project contracts in construction works

b Explain in detail the importance of project cost management in construction works CO2

Roll No. | | | | | | | |

Mid Semester Examination 2022

Name of the Course: M Tech (Geo-Informatics) Semester: III

Name of the Paper: Application of Geo-informatics for Natural Resources Management

Paper Code: MGI 313

Time: 1:1/2 Hours Maximum Marks: 50

**Note:-**

(v) All **the questions** by choosing **any one of the sub questions.**

(vi) Each question carries 10 marks

| | | |
|---|---|---|
| **Q1** | (10marks) | **CO1** |
| (a) | Brief note on soil mapping and soil classification using remote sensing | |
| (b) | Write briefly on the genetic classification of drainage type define important drainage patterns with a neat diagram. | |
| **Q2** | (10 marks) | **CO1** |
| (a) | Define the following landforms: (a) Glacial landforms with diagram (b) Fracture, Fault and Joints. | |
| (b) | Brief note on Geomorphological Mapping | |
| **Q3** | (10 marks) | **CO2** |
| (a) | Write a brief note on the genetic classification of drainage type and define the important drainage patterns with the neat diagram. | |
| (b) | Brief note on Soil salinity mapping using Remote sensing | |
| **Q4** | (10 marks) | **CO2** |
| (a) | Write short notes on the following:<br>(a) Reconnaissance Survey<br>(b) Detailed survey. | |
| (b) | Differentiate the following: soil salinity mapping and soil erosion mapping. | |
| **Q5** | (10 marks) | **CO2** |
| (a) | Explain the method and stages of soil survey. | |
| (b) | Brief note on Mineral resources exploration and Mineral mapping using RS & GIS | |

Roll No. | | |

Roll No.

# MID Semester Examination 2022

Name of the Course:M.Tech | Semester: 3rd

Name of the Paper: Soil Structure Interaction | Paper Code: **MSE 315**

Time: 1.5 Hours | Maximum Marks: 50

Note:-

(i) All questions are compulsory.

(ii) Answer any two sub questions among a & b in each main question

(iii) Total marks in each main question are twenty.

(iv) Each question carries 10 marks

**Q1** (10marks)

(a) Derive an expression for equation of motion for flexible base..

(b) Explain effect of soil structure interaction for a structure on soft soil with the help of a neat diagram. **CO1**

**Q2** (10 marks)

(a) What are the various principle problems in soil structure interaction.

(b) Explain types of prescribed loadings in soil structure interaction due to seismic excitation. **CO1**

**Q3** (10 marks)

(a) Explain effect of soil structure interaction for a structure on hard rocky ground with the help of a neat diagram.

(b) Explain in brief about objectives and significance of soil structure interaction **CO1**

**Q4** (10 marks)

(a) Derive an expression for equation of motion for rigid base.

(b) Distinguish between continuous model, discrete model, and finite element model in context to soil structure interaction. **CO2**

**Q5** (10 marks)

(a) Explain Frequency content and spatial variation of applied loads.

(b) Explain in brief about direct and sub structure methods. **CO2**

Roll No. | | | | | | | |

Paper Code: OCE-351

M.Tech. (Structural Engg)

**Mid Semester Examination 2022**

III *Semester*

**Cost Management of Engineering Projects**

Time: 1:30 Hours

MM: 50

**Note:**

(i) All questions are compulsory.

(ii) Answer any one sub questions among a & b in each main question.

(iii) Total marks for each main question are ten.

Q1 (1x10 = 10 Marks)

a Describe Opportunity cost, Incremental cost, Differential cost and Relevant cost.

b Explain in detail Cost concepts in decision-making. CO1

Q2 (1x10 = 10 Marks)

a Explain Creation of a Database for operational control?

b Describe the objectives of a costing system? CO1

Q3 (1x10 = 10 Marks)

a Describe project execution as conglomeration of technical and nontechnical activities.

b Describe briefly pre-project execution and its importance in project work. CO2

Q4 (1x10 = 10 Marks)

a Describe briefly pre project execution main clearances and its impact on project.

b Explain in detail the importance of project site and project documentation phases. CO2

Q5 (1x10 = 10 Marks)

a Explain in detail project contracts in construction works

b Explain in detail the importance of project cost management in construction works CO2

Mid Semester Examination 2022

M.Tech. (Structural Engineering)
Design of Bridges

*Semester: III Sem*
*Course Code:* MSE 328

Time: 1-1/2 Hour

Maximum Marks: 50

Note:
(i) Answer all the questions by choosing *any one of the sub questions.*
(ii) Each question carries 10 marks

Q1

(a) Discuss the different types of bridges. (10 marks) CO1

OR

(b) Describe the process for site selection of bridges.

Q2

(a) Explain a reinforced concrete bridge and its components with the help of a neat diagram. (10 marks) CO1

OR

(b) Explain a cable suspension suspension bridge and its components with the help of a neat diagram

Q3

(a) Explain the IRC class loading for wheeled and tracked vehicle of IRC class AA and class 70R vehicle. (10 marks) CO1

OR

(b) Explain the IRC class loading for IRC class A and class B vehicle.

Q4

(a) Analyse and draw SFD and BMD for a reinforced concrete slab culvert for National Highway crossing for following data:
Carriageway: 2 lane -7.5m wide, clear span= 6m , width of bearing= 400mm , loading- IRC class AA tracked vehicle. (10 marks) CO2

OR

(b) Analyse and draw SFD and BMD for a reinforced concrete slab culvert for National Highway crossing for following data:
Carriageway: 1 lane - 4m wide, clear span= 7m , width of bearing= 400mm , loading- IRC class 70R wheeled vehicle.

Q5

(a) Design a reinforced concrete slab culvert for National Highway crossing for following data:
Carriageway: 2 lane -7.5m wide, clear span= 6m , width of bearing= 400mm , loading- IRC class AA tracked vehicle. M25 concrete and Fe500 steel. (10 marks) CO2

OR

(b) Design a reinforced concrete slab culvert for National Highway crossingfor following data:
Carriageway: 1 lane - 4m wide, clear span= 7m , width of bearing= 400mm , loading- IRC class 70R wheeled vehicle. M25 concrete and Fe500 steel.

Roll No.....

Mid Semester Examination 2022

*Name of the Program*: M.Tech. (CSE) *Semester*: 3rd

*Name of the Course*: Intrusion Detection Systems *Course Code*: **MCS 324**

Time: 90 Minutes Maximum Marks: 50

**Note:**

(i) Answer **all the questions** by choosing ***any one of the sub questions***.

(ii) Each question carries 10 marks

| | | |
|---|---|---|
| **Q1** | (10 marks) | |
| (a) | What do you understand about network taxonomies? How Hansman's taxonomy is more complete and practical. Explain it. | CO1 |
| | OR | |
| (b) | What is Payload attack? Classify and define the categories of Payload attack | |
| **Q2** | (10 marks) | |
| (a) | What is Security Testing? List the attributes of Security Testing and explain them? | |
| | OR | CO1 |
| (b) | What do you understand about Buffer Overflow Attacks, Misconfiguration Attacks and Man-in-the-Middle Attacks? Discuss about these attacks with suitable example. | |
| **Q3** | (10 marks) | |
| (a) | Discuss about miss-use detection with suitable diagram? Is this attack suitable for fully effective in uncovering unknown attacks or known attacks? | CO2 |
| | OR | |
| (b) | Classify and discuss various techniques which are commonly used to implement misuse detection. | |
| **Q4** | (10 marks) | |
| (a) | Discuss about a typical anomaly detection system with suitable diagram. Name the various models available in anomaly detection system. Explain Advanced Statistical Models in details. | CO2, |
| | OR | |
| (b) | What are the important sources that is used for data collection for development of Host based Intrusion detection systems? | |
| **Q5** | (10 marks) | |
| (a) | Are specification-based detection approaches based on misuse or anomaly based? Discuss your answers, is it both or none of these? | CO2, CO3 |
| | OR | |
| (b) | Discuss about Supervised and Unsupervised Anomaly Detection techniques. | |

Mid Semester Examination 2022

*Name of the Program*: **M.Tech.(CAD/CAM)** *Semester:* III

*Name of the Course*: **Cost Management of Engineering Projects**

*Course Code:* **OEME02**

**Time: 1 1/2 Hour** **Maximum Marks: 50**

**Note:**

(i) Answer **all the questions** by choosing ***any one of the sub questions***.

(ii) Each question carries 10 marks

| | | |
|---|---|---|
| Q1 | (10 marks) | **CO1** |
| a) | Describe the importance of cost management of engineering projects. | |
| | **OR** | |
| b) | What are the objectives of costing system? How it is helpful in decision making? | |
| Q2 | (10 marks) | **CO1** |
| a) | What do you mean by Management? Also explain the role of budget planning for controlling the completion of projects on time. | |
| | **OR** | |
| b) | What is the cost? Explain the cost concepts in decision making.. | |
| Q3 | (10 marks) | **CO1 ,CO2** |
| a) | Explain the differential cost with example. | |
| | **OR** | |
| b) | Explain the incremental cost and opportunity cost. | |
| Q4 | (10 marks) | **CO2** |
| a) | Write the different stages of project. | |
| | **OR** | |
| b) | What is the importance of projects site? | |
| Q5 | (10 marks) | **CO2 , CO3** |
| a) | How the members of projects team are responsible for completion of projects on time? | |
| | **OR** | |
| b) | What do you understand by commissioning of projects? | |

Mid SemesterExamination 2022

*Name of the Program*: M.Tech *Semester*: III

*Name of theCourse*:Computer Aided Analysis of Mechanical Systems *Course Code*:**DEME53**

Time: 1 1/2 Hour Maximum Marks: 50

**Note:**

(i) Answer **all the questions** by choosing ***any one of the sub questions***.

(ii) Each question carries 10 marks

| | | |
|---|---|---|
| Q1 | (10 marks) | **CO1** |
| a) | Explain the various types of mechanical systems whose analysis can be possibly carries out using a computer? | |
| | **OR** | |
| b) | Why is it important to use a computer to analyze a mechanical system? | |
| Q2 | (10 marks) | **CO1** |
| a) | Derive the governing equation of a single degree of freedom spring mass damper system and write is as a system of first order ordinary differential equations. | |
| | **OR** | |
| b) | Derive the governing equation of a two degree of freedom quarter car modeland write is as a system of first order ordinary differential equations. | |
| Q3 | (10 marks) | **CO1 and CO2** |
| a) | Explain the finite difference method using Taylor series. | |
| | **OR** | |
| b) | Explain Euler integration technique in detail. | |
| Q4 | (10 marks) | **CO2** |
| a) | How can you apply RK2 technique to a single degree of freedom quarter car model? | |
| | **OR** | |
| b) | How can you apply RK4 technique to a spring mass damper? | |
| Q5 | (10 marks) | **CO2** |
| a) | With help of a diagram derive the governing partial differential equation for a string. | |
| | **OR** | |
| b) | What is a boundary value problem? Give one example. | |

**Note for the question paper setters:**

- **Question paper should have questions from each unit and cover all COs**
- **Sub questions (a) and (b) can be further divided if required**

Roll No.....

# Mid Semester Examination 2022

*Name of the Program*: M.Tech (VLSI Design) *Semester: 3rd*
*Name of the Course*:Internet of Things *Course Code:* VOM 301

Time: 90 Minutes Maximum Marks: 50

**Note:**

(i) Answer **all the questions** by choosing ***any one of the sub questions***.
(ii) Each question carries 10 marks

| | | |
|---|---|---|
| **Q1** | (10 marks) | |
| (a) | **Explain overview and motivation for IoT.** | CO 1 |
| | OR | |
| (b) | Explain IoT Applications in details. | |
| **Q2** | (10 marks) | |
| (a) | **Define IoT Architecture. Use diagrams in support of your answer.** | CO1 |
| | OR | |
| (b) | **What are basic nodal capabilities in context of IoT.** | |
| **Q3** | (10 marks) | |
| (a) | **Explain objects and services for IoT.** | CO2 |
| | OR | |
| (b) | **Explain RFID Technology in details.** | |
| **Q4** | (10 marks) | |
| (a) | **Explain role of sensor technology in context of IoT.** | CO2 |
| | OR | |
| (b) | **How security and privacy plays role in context of IoT** | |
| **Q5** | (10 marks) | |
| (a) | **Define IoT Framework.** | CO1 |
| | OR | |
| (b) | **Explain Interoperability.** | CO2 |

Roll No.....

# Mid Semester Examination 2022

*Name of the Program*: M.Tech (VLSI Design) *Semester*: 3rd

*Name of the Course*:Internet of Things *Course Code*: VOM 301

Time: 90 Minutes Maximum Marks: 50

**Note:**

(i) Answer **all the questions** by choosing ***any one of the sub questions***.

(ii) Each question carries 10 marks

| | | |
|---|---|---|
| Q1 | (10 marks) | CO 1 |
| (a) | **Explain overview and motivation for IoT.** | |
| | OR | |
| (b) | Explain IoT Applications in details. | |
| Q2 | (10 marks) | CO1 |
| (a) | **Define IoT Architecture. Use diagrams in support of your answer.** | |
| | OR | |
| (b) | **What are basic nodal capabilities in context of IoT.** | |
| Q3 | (10 marks) | CO2 |
| (a) | **Explain objects and services for IoT.** | |
| | OR | |
| (b) | **Explain RFID Technology in details.** | |
| Q4 | (10 marks) | CO2 |
| (a) | **Explain role of sensor technology in context of IoT.** | |
| | OR | |
| (b) | **How security and privacy plays role in context of IoT** | |
| Q5 | (10 marks) | CO1 |
| (a) | **Define IoT Framework.** | |
| | OR | |
| (b) | **Explain Interoperability.** | CO2 |

Roll No.....

# Mid Semester Examination

*Name of the Program*: M.Tech (VLSI Design and Systems) *Semester:* IIIrd
*Name of the Course*: Organic Electronic Devices and Circuits *Course Code:* VDM 391

Time: 90 Minutes Maximum Marks: 50

**Note:**

(i) Answer **all the questions** by choosing ***any one of the sub questions***.
(ii) Each question carries 10 marks

| | | |
|---|---|---|
| **Q1** | (10 marks) | **CO1** |
| (a) | Describe the differences between small molecules and conductive polymers with the help of a suitable example. | |
| | OR | |
| (b) | Why are HOMO and LUMO energy states important? and describe the movement of charges within an organic semiconductor. | |
| **Q2** | (10 marks) | **CO1** |
| (a) | Explain the various organic semiconductor, contact electrode, dielectric and substrate material along with their names for developing Organic Thin Film Transistor (OTFT). | |
| | OR | |
| (b) | Clarify the operating principle of Organic Thin Film Transistor (OTFT) with a neat suitable diagram and draw output and transfer characteristics with a suitable diagram. | |
| **Q3** | (10 marks) | **CO1/CO2** |
| (a) | Describe the single-gate (SG) and double-gate (DG) OTFT structures. | |
| | OR | |
| (b) | Describe the various applications of OTFT and describe the merits and demerits of OTFTs over conventional MOSFETs. | |
| **Q4** | (10 marks) | **CO1/CO2** |
| (a) | Draw chemical structure of following materials:<br>(i)Pentacene (ii) Tetracene<br>(ii)TIPS Pentacene (iv) P3HT<br>(iii) Poly-acetylene (vi) poly-thiophene | |
| | OR | |
| (b) | Why bottom gate OTFT is preferred over top gate OTFT, explain and describe the flow diagram of the fabrication process for OTFT in top and bottom contact structure. | |
| **Q5** | (10 marks) | **CO1/CO2** |
| (a) | Describe the Merits and demerits of organic Semiconductor over Silicon. | |
| | OR | |
| (b) | Describe the basic Fabrication Steps of SG and DG of OTFT. | |

**Note for the question paper setters: (Assuming two units are covered)**

- **Question paper should have questions from both the units covering the related COs**
- **Sub questions (a) and (b) can be further divided if required**

Mid Semester Examination 2022

*Name of the Program*: **M.Tech CSE**

*Semester:* 1st

*Name of the Course*: **Soft Computing**

*Course Code:* **MCS301**

Time: 1 Hour

Maximum Marks: 50

**Note:**

(i) Answer **all the questions** by choosing ***any one of the sub-questions***.

(ii) Each question carries 10 marks

| | | |
|---|---|---|
| Q1 | (10 marks) | CO1 |
| a) | Compare and contrast biological neurons and artificial neurons | |
| | **OR** | |
| b) | With suitable architecture write the training algorithm used in ADALINE network | |
| Q2 | (10 marks) | CO1 |
| a) | Define fuzzification? Discuss any two methods of assigning membership values? | |
| | **OR** | |
| b) | "Approximate reasoning is important in fuzzy logic"- Justify the statement | |
| | | |
| Q3 | (10 marks) | CO2 |
| a) | Justify XOR function is non Linearly separability separable by a single decision boundary line | |
| | **OR** | |
| b) | Implement AND function using McCulloch-Pitts Neuron (take binary data). | |
| Q4 | (10 marks) | CO3 |
| a) | Implement a perceptron network for an OR function with bipolar inputs and targets. Assume necessary parameters. Perform one epoch of training | |
| | **OR** | |
| b) | Using the Hebb rule, find the weights required to perform the following classification of the given input pattern. The pattern is shown as 3*3 matrix form in the squares. The "+" symbol represents the value "1" and empty squares indicate "-1". Consider "I" belongs to the member of the class (so has target value 1) and "O" does not belong to the member of class (so has target value -1). | |
| Q5 | (10 marks) | CO3 |
| a) | For the network shown in figure, calculate the net input to the output neuron | |

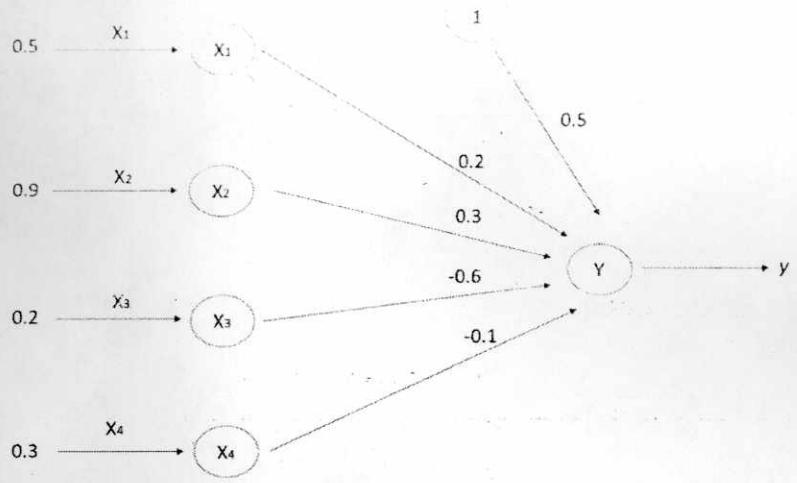
| | | |
|---|---|---|
| + | + | + |
| | + | |
| + | + | + |

'I'

| | | |
|---|---|---|
| + | + | + |
| + | | + |
| + | + | + |

'O'

**Data for Input patterns**




**OR**

b) Consider two fuzzy sets

$A=\{\frac{1}{2.0}+\frac{0.65}{4.0}+\frac{0.5}{6.0}+\frac{0.35}{8.0}+\frac{0}{10.0}\}$ $B= A=\{\frac{0.2}{2.0}+\frac{0.55}{4.0}+\frac{0.7}{6.0}+\frac{0.65}{8.0}+\frac{1}{10.0}\}$

Find the following

(1) AUB (2) A∩B (3) AU$\bar{A}$ (4) $\overline{AUB}$ (5) $\overline{A \cap B}$